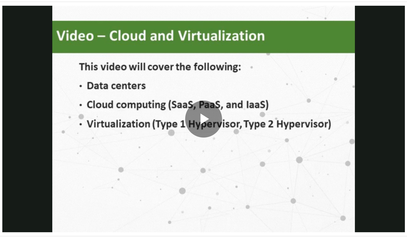
Cloud Computing

13.1.1

Video - Cloud and Virtualization

Click Play for an overview of cloud computing and virtualization.



13.1.2

Cloud Overview

In the previous video, an overview of cloud computing was explained. Cloud computing involves large numbers of computers connected through a network that can be physically located anywhere. Providers rely heavily on virtualization to deliver their cloud computing services. Cloud computing can reduce operational costs by using resources more efficiently. Cloud computing addresses a variety of data management issues:

- Enables access to organizational data anywhere and at any time
- Streamlines the organization's IT operations by subscribing only to needed services
- Eliminates or reduces the need for onsite IT equipment, maintenance, and management
- Reduces cost for equipment, energy, physical plant requirements, and personnel training needs
- Enables rapid responses to increasing data volume requirements

Cloud computing, with its "pay-as-you-go" model, allows organizations to treat computing and storage expenses more as a utility rather than investing in infrastructure. Capital expenditures are transformed into operating expenditures.

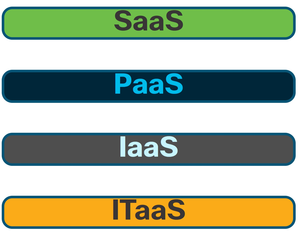
13.1.3

Cloud Services

Cloud services are available in a variety of options, tailored to meet customer requirements. The three main cloud computing services defined by the National Institute of Standards and Technology (NIST) in their Special Publication 800–145 are as follows:

- **Software as a Service (SaaS)** – The cloud provider is responsible for access to applications and services, such as email, communication, and Office 365 that are delivered over the Internet. The user does not manage any aspect of the cloud services except for limited user–specific application settings. The user only needs to provide their data.
- **Platform as a Service (PaaS)** – The cloud provider is responsible for providing users access to the development tools and services used to deliver the applications. These users are typically programmers and may have control over the configuration settings of the cloud provider's application hosting environment.
- **Infrastructure as a Service (IaaS)** – The cloud provider is responsible for giving IT managers access to the network equipment, virtualized network services, and supporting network infrastructure. Using this cloud service allows IT managers to deploy and run software code, which can include operating systems and applications.

Cloud service providers have extended this model to also provide IT support for each of the cloud computing services (ITaaS), as shown in the figure. For businesses, ITaaS can extend the capability of the network without requiring investment in new infrastructure, training new personnel, or licensing new software. These services are available on demand and delivered economically to any device anywhere in the world without compromising security or function.

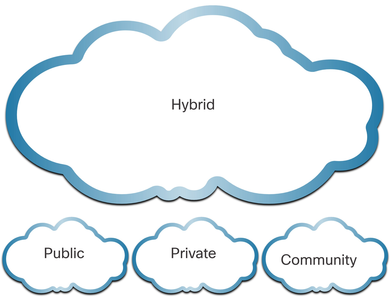


13.1.4

Cloud Models

There are four primary cloud models, as shown in the figure.

- **Public clouds** – Cloud-based applications and services offered in a public cloud are made available to the general population. Services may be free or are offered on a pay-per-use model, such as paying for online storage. The public cloud uses the Internet to provide services.
- **Private clouds** – Cloud-based applications and services offered in a private cloud are intended for a specific organization or entity, such as the government. A private cloud can be set up using the organization's private network, though this can be expensive to build and maintain. A private cloud can also be managed by an outside organization with strict access security.
- **Hybrid clouds** – A hybrid cloud is made up of two or more clouds (example: part private, part public), where each part remains a separate object, but both are connected using a single architecture. Individuals on a hybrid cloud would be able to have degrees of access to various services based on user access rights.
- **Community clouds** – A community cloud is created for exclusive use by a specific community. The differences between public clouds and community clouds are the functional needs that have been customized for the community. For example, healthcare organizations must remain compliant with policies and laws (e.g., HIPAA) that require special authentication and confidentiality.



13.1.5

Cloud Computing versus Data Center

The terms data center and cloud computing are often used incorrectly. These are the correct definitions of data center and cloud computing:

- **Data center**: Typically, a data storage and processing facility run by an in-house IT department or leased offsite.
- **Cloud computing**: Typically, an off-premise service that offers on-demand access to a shared pool of configurable computing resources. These resources can be rapidly provisioned and released with minimal management effort.

Data centers are the physical facilities that provide the compute, network, and storage needs of cloud computing services. Cloud service providers use data centers to host their cloud services and cloud-based resources.

A data center can occupy one room of a building, one or more floors, or an entire building. Data centers are typically very expensive to build and maintain. For this reason, only large organizations use privately built data centers to house their data and provide services to users. Smaller organizations that cannot afford to maintain their own private data center can reduce the overall cost of ownership by leasing server and storage services from a larger data center organization in the cloud.

13.1.6

Check Your Understanding - Cloud Computing

Check your understanding of cloud computing by choosing the BEST answer to the following questions.

1. Which cloud service provides programmers access to the development tool and services used to deliver applications?
   - SaaS
   - PaaS
   - IaaS

2. Which cloud service provides users with access to services such as email, communications, and Office 365 delivered over the internet?
   - SaaS
   - PaaS
   - IaaS

3. Which cloud service provides IT managers with access to network equipment, virtualized services, and a supporting network infrastructure?
   - SaaS
   - PaaS
   - IaaS

4. Which cloud model represents two or more clouds where each part remains a distinctive object, but both are connected using a single architecture?
   - public cloud
   - private cloud
   - hybrid cloud
   - community cloud

5. Which cloud model is used to meet the needs of a specific industry, such as healthcare or media?
   - public cloud
   - private cloud
   - hybrid cloud
   - community cloud

Check

Show Me

Reset